

PAPER Direct Energy Conversion for a D–³He Tandem-Mirror Burner: Recovering Charged-Particle Power Directly Without a Thermal Cycle

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Abstract

A deuterium–helium-3 (D–³He) tandem-mirror burner puts most of its fusion power into charged particles, so it admits an option a tokamak does not: recover the exhaust electrostatically, as electricity, without a thermal cycle or a turbine on the direct channel. This paper is the flagship treatment of that option for the Kronos burner (engineering gain $Q_E = 1.318$ at $P_{\text{fus}} = 4298.5$ MW, neutron fraction $f_n = 5.44\%$, plug potential $e\phi_i = 248.75$ keV, on the open field lines downstream of a 26.49 T plug coil that is a distinct magnet from the breeder’s 16.84 T centrepost). We write the governing physics as explicit equations: the Pastukhov ambipolar confinement that sets the axial end-loss channel; the Stix collisional slowing-down partition, which shows that at a reactor electron temperature $T_e \approx 89.7$ keV the 14.68 MeV fusion proton is born more than an order of magnitude above its critical energy and hands $\approx 83\%$ of its energy to the electrons, so that the thermalized axial exit carries an *electron-dominant* split ($\approx 55.5\%$ electron/thermal against $\approx 44.5\%$ directed ion) that inverts the received birth-spectrum picture and holds across 20–200 keV; the multi-stage electrostatic collector and its space-charge limit; the travelling-wave converter for the proton channel; and a quasineutral two-species expander that recovers the directed electron stream against the ambipolar potential of the expansion. We resolve the direct-converter efficiency into a defensible ladder — frozen conservative anchor $\eta_{\text{dec}} = 0.49$, floor 0.56, Moir–Barr design 0.70, ideal proton-only ceiling 0.85 — and place it correctly in the plant balance: booked at the design value the burner delivers $P_{\text{out}} = 3522.0$ MWe against $P_{\text{in}} = 2672.0$ MWe (+850.0 MWe net, recirculating fraction 0.759), but the net-positive knee sits at $\eta_{\text{dec}} \approx 0.52$, so *conversion is a bounded upside lever, not the closure mechanism* — closure is gated by the recirculating plug power, and central-cell length, not converter efficiency, buys the net margin (+104/ + 850/ + 2832 MWe at 55/440/1400 m). We give the GPU-HPC methodology (the codes, not dollars), map the converter onto the two products — Aegis forward-base (grid-independent black-start in ≈ 42 s) and MetroVolt data-center (HVDC-native 88.1 % source-to-chip against 75.9 % for a conventional AC chain) — and benchmark against the published venetian-blind, travelling-wave and expander literature. Kronos has built no converter; every headline quantity is a frozen anchor in the de-risking register^[R0].

Keywords: direct energy conversion; deuterium–helium-3; tandem mirror; ambipolar potential; Pastukhov confinement; travelling-wave converter; quasineutral expander; HVDC-native power; advanced-fuel fusion

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Nomenclature

Symbol	Definition
E_p, E_α	birth energy of the D– ³ He proton (14.68 MeV) and ⁴ He (3.67 MeV)
E_c	Stix critical energy (fast-ion equipartition between background ions and electrons)
f_e	fraction of fast-ion birth energy delivered to electrons during slowing-down
f_n, f_{dir}	neutron fraction of fusion power (5.44 %); charged fraction of exhaust (0.4945)
$e\phi_i$	ambipolar (plug) potential barrier confining central-cell ions
T_i, T_e	central-cell ion and electron temperature
$R_{\text{mc}}, R_{\text{exp}}$	plug mirror ratio; magnetic-expansion ratio of the end expander
$\tau_s, \tau_{\parallel}$	collisional slowing-down time; Pastukhov axial-confinement time
$\eta_{\text{dec}}, \eta_{\text{th}}, \eta_{\text{grid}}$	direct-converter efficiency; thermal-cycle efficiency; grid transparency factor
$P_{\text{fus}}, P_{\text{ax}}, P_{\text{exh}}$	fusion power; charged axial end-loss power; total exhaust power
$P_{\text{out}}, P_{\text{in}}$	gross electric output; recirculating electric input (heating/plug drive)
Q_E	engineering (net-electric) gain, $P_{\text{out}}/P_{\text{in}}$
J_{CL}	Child–Langmuir space-charge-limited current density

1 Introduction

1.1 Why direct energy conversion, and why here

A magnetically confined fusion reactor that burns a deuterium–tritium (D–T) plasma puts $\approx 80\%$ of its yield into 14.1 MeV neutrons, which can only be captured as heat and run through a Rankine or Brayton cycle; the conversion to electricity is Carnot-bounded, and the machine inherits the whole thermal-plant apparatus. A deuterium–helium-3 burner is different in kind. The primary reaction



releases its energy entirely in charged particles, and even with the parasitic D–D and D–T side reactions the neutron fraction of total fusion power at the Kronos operating point is only $f_n = 5.44\%$ [1]. A stream of charged particles can, in principle, be decelerated in an electrostatic field and delivered directly as electricity — *direct energy conversion* (DEC) — at an efficiency set by collector physics rather than by Carnot, and with the turbine hall deleted for the direct channel [2, 5]. This is the single largest structural advantage of the advanced-fuel line, and it is why the Kronos burner is an open-ended linear device — a tandem mirror with a direct converter on the fanned-out field lines at each end — rather than a torus.

The idea is old: Post proposed periodic electrostatic and expander converters for mirror machines [4, 5], and Moir and Barr built and analysed the venetian-blind converter for D–T and advanced fuels, reaching $\approx 70\%$ collector efficiency in design [2, 3]. What has changed is that a compact, high-field D–³He tandem mirror is now a concrete design point rather than a fuel-cycle sketch, so the converter can — and must — be specified against a real exhaust spectrum, a real recirculating power balance, and a real product. That is the purpose of this paper.

1.2 The tandem-mirror burner in one paragraph

The Kronos burner confines a long central cell ($\ell_c = 440 \text{ m}$ at the ratified design unit) between two high-field end plugs [1, 39]. An ambipolar electrostatic potential $e\phi_i$, generated at the plugs, exponentially enhances the axial confinement of central-cell ions (Pastukhov confinement [6, 7, 8]); at the design point $e\phi_i = 248.75 \text{ keV}$ with $\phi_i/T_i = 2.764$, and this exponential is what closes the burn. The plug field is produced by a no-insulation REBCO coil at a peak field of 26.49 T — a distinct, higher-field magnet from the breeder’s 16.84 T centrepost, and the two must never be conflated [32, 1]. Ions that do leak over the barrier, together with the 14.68 MeV fusion protons, flow out along the open field lines, fan out in a magnetic *expander*, and strike the direct converter. This paper takes the burner’s frozen 0-D power balance as given (companion papers close it [30, 31, 33]) and asks the DEC question: what is actually in that exhaust, how is it recovered, at what efficiency, and how much does it matter to the plant.

1.3 Contributions

Four contributions follow. (i) We write the governing physics of the burner’s DEC problem as explicit equations (§2): the Pastukhov end-loss channel, the Stix slowing-down partition, the electrostatic-collector deceleration and its space-charge limit, the travelling-wave converter, the quasineutral expander, and the plant power balance. (ii) We establish, from the slowing-down partition, that the dominant axial-loss channel at reactor temperature is a directed *electron* stream, not the 14.68 MeV proton — an inversion of the received picture that reframes what the converter

must recover. (iii) We resolve the converter efficiency into a defensible ladder and place it correctly in the recirculating balance, showing that DEC is a bounded upside lever whose net knee sits just above the frozen conservative anchor, so that closure is gated by the plug power, not by conversion. (iv) We give the GPU-HPC methodology and map the converter onto the two products, Aegis and MetroVolt, before benchmarking against the published literature (§5). Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register [1]; serial identifiers (e.g. BU-L1-A10) are given inline so each number is auditable.

2 Governing equations and the formal problem

2.1 $D-^3\text{He}$ energetics and the charged-power fraction

The primary reaction (1) shares its 18.35 MeV between the two charged products by momentum conservation, giving the proton

$$\frac{E_p}{E_p + E_\alpha} = \frac{14.68}{18.35} = 0.800 \quad (2)$$

of the charged birth energy (figure 1a). The volumetric fusion power for a Maxwellian central cell of ion densities n_D, n_{He3} is

$$P_{\text{fus}} = n_D n_{\text{He3}} \langle \sigma v \rangle_{D^3\text{He}}(T_i) Q + P_{\text{DD}} + P_{\text{DT}}, \quad (3)$$

with the reactivity $\langle \sigma v \rangle$ taken from the Bosch–Hale parameterization [15]. The D–D branch and the D–T reaction on the small bred-tritium inventory supply the only neutrons; at the Kronos operating point the neutron fraction is

$$f_n \equiv \frac{P_n}{P_{\text{fus}}} = \frac{233.95 \text{ MW}}{4298.5 \text{ MW}} = 0.0544, \quad (4)$$

so 94.56 % of the fusion power leaves as charged particles (figure 1b) — the raw material for direct conversion, and the reason the $D-^3\text{He}$ line can site a burner where a D–T machine cannot [17, 16].

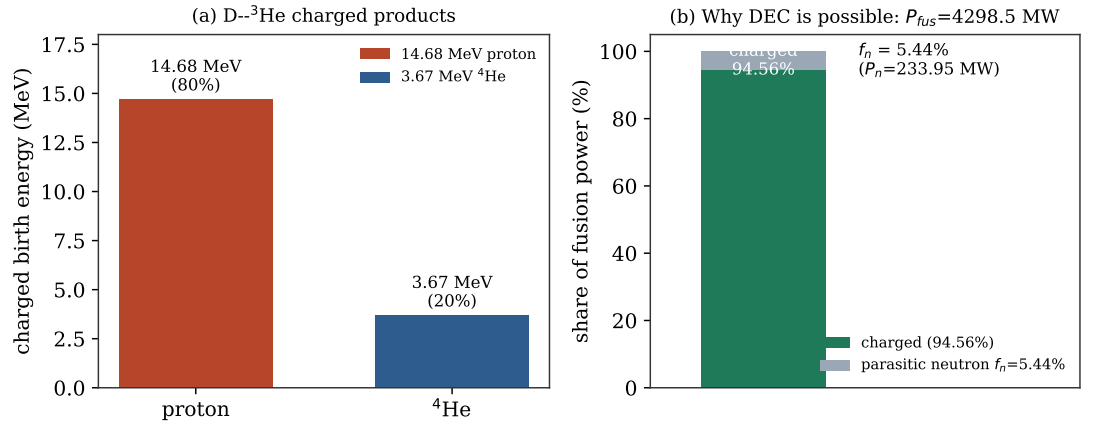


Figure 1. $D-^3\text{He}$ energetics. **(a)** The 14.68 MeV proton carries 80 % of the charged birth energy. **(b)** At the Kronos operating point only $f_n = 5.44\%$ of fusion power ($P_{\text{fus}} = 4298.5 \text{ MW}$) is neutron; the charged remainder is the direct-conversion feedstock. Values are register anchors (BU-L1-A1) [1].

2.2 The end-loss channel: Pastukhov ambipolar confinement

Central-cell ions are confined not only by the mirror ratio R_{mc} but by the electrostatic barrier $e\phi_i$ raised at the plugs. For a species confined by a potential ϕ in a mirror of ratio R , the collisional (Pastukhov) axial-confinement time, in the Cohen–Rensink corrected form [6, 7], scales as

$$\tau_{\parallel} \simeq \tau_s g(R) \frac{e\phi_i}{T_i} \exp\left(\frac{e\phi_i}{T_i}\right) / \ln\left(\frac{R+1}{1}\right), \quad (5)$$

where τ_s is the ion–ion collision time and $g(R)$ a slowly varying $O(1)$ mirror function. The exponential in (5) is the whole game: at $\phi_i/T_i = 2.764$ the barrier multiplies the confinement by more than an order of magnitude, which is why the plug potential — not the converter — is the burner’s binding lever [1, 33]. An ion that does escape crosses the barrier and arrives at the

expander with a directed axial energy of order $e\phi_i$ plus a few T_i , populating a central-cell end-loss spectrum in the 253–513 keV window (figure 4); the 14.68 MeV fusion protons form a separate, monoenergetic high channel. These two populations are what the converter train must sort and decelerate.

2.3 The Stix slowing-down partition: the loss channel is a directed electron stream

The received framing of D–³He DEC points at the 14.68 MeV proton and an ion converter, because the proton carries most of the *birth* energy. That framing describes the birth spectrum, not the axial loss. The fate of a fast ion is set by where it sits relative to the Stix critical energy E_c — the energy at which it sheds equal power to background ions and electrons [11]:

$$E_c \simeq 14.8 T_e \left(\frac{\bar{A}^{2/3}}{Z_{\text{eff}}} \right), \quad (6)$$

with $\bar{A}$ the background mean mass and Z_{eff} the effective charge. At $T_e = 89.72$ keV (the synchrotron-pinned operating temperature [1]) $E_c \approx 1.33$ MeV, so the 14.68 MeV proton is born at $E_0/E_c \approx 11$ — more than an order of magnitude above critical — and slows predominantly on the electrons. The fraction of a fast ion’s energy delivered to electrons during collisional slowing-down from E_0 follows from the Fokker–Planck drag [12, 13],

$$f_e(E_0) = 1 - \frac{E_c}{E_0} \int_0^{E_0/E_c} \frac{dy}{1 + y^{3/2}}, \quad (7)$$

which for the proton at $E_0/E_c \approx 11$ gives $f_e \approx 0.83$ (figure 2): most of the proton’s birth energy is handed to the electrons before it can be collected as a directed ion. Booking (7) against the finite axial-confinement time $\tau_{\parallel}$ of (5) — the ion leaves the mirror before it fully thermalizes — yields the *thermalized* split at the axial exit: the electron/thermal stream is $\approx 55.5\%$ and the residual directed-ion stream $\approx 44.5\%$ (figure 3a; frozen anchor [1]). The electron channel is the larger one. Because $E_c \propto T_e$, raising the temperature lowers E_0/E_c and softens the effect, but the electron share stays above one-half across the entire reactor-relevant range 20–200 keV (figure 3b): *the inversion is robust*, and the DEC problem is, at root, an electron-recovery problem. No published converter recovers a directed electron stream — hot-emitter collectors flood their own gap through the Richardson–Dushman back-emission wall, and ion converters treat electrons as a suppressible loss — which is the open problem §2.5 addresses.

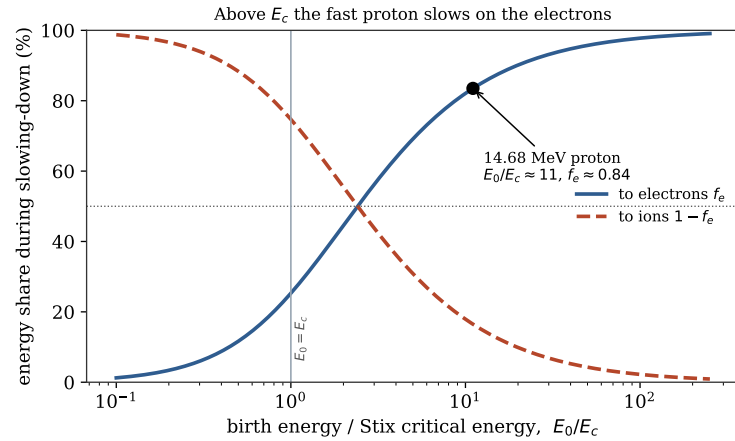


Figure 2. The Stix slowing-down partition, equation (7). Above the critical energy E_c a fast ion sheds its energy preferentially to the electrons; the 14.68 MeV proton at $T_e = 89.7$ keV sits at $E_0/E_c \approx 11$ and delivers $f_e \approx 0.83$ of its energy to the electron channel.

2.4 Electrostatic-collector deceleration and the space-charge limit

The directed channel is recovered by decelerating each particle against a graded electrostatic potential. A particle of energy E entering an N -stage collector is captured on the electrode whose retarding potential eV_k satisfies $eV_k \leq E < eV_{k+1}$; it delivers eV_k to the circuit and dumps the

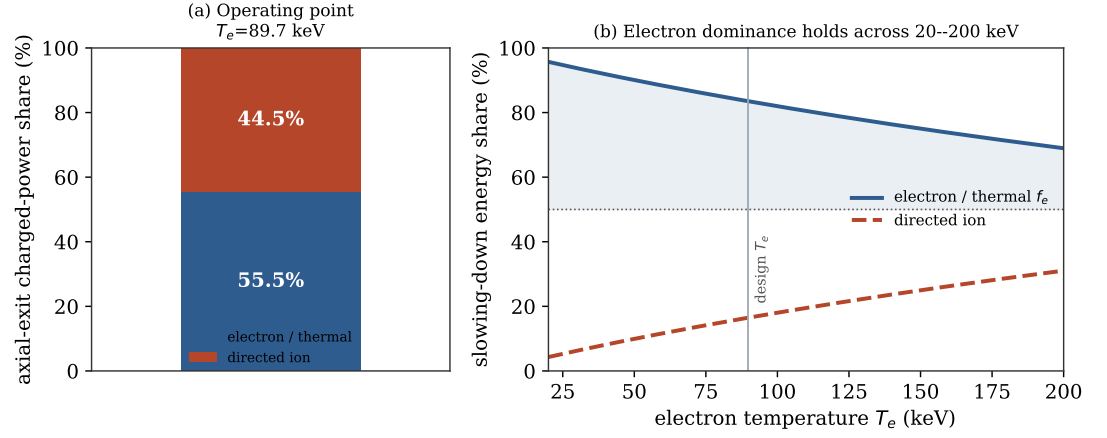


Figure 3. The electron-dominant axial exit. **(a)** Thermalized split at the operating point: electron/thermal 55.5% against directed-ion 44.5% (frozen anchor [1]). **(b)** The electron share (computed from equation (7) with $E_c \propto T_e$) stays above 50% across 20–200 keV: the inversion of the birth-spectrum picture is robust across the reactor-relevant temperature range.

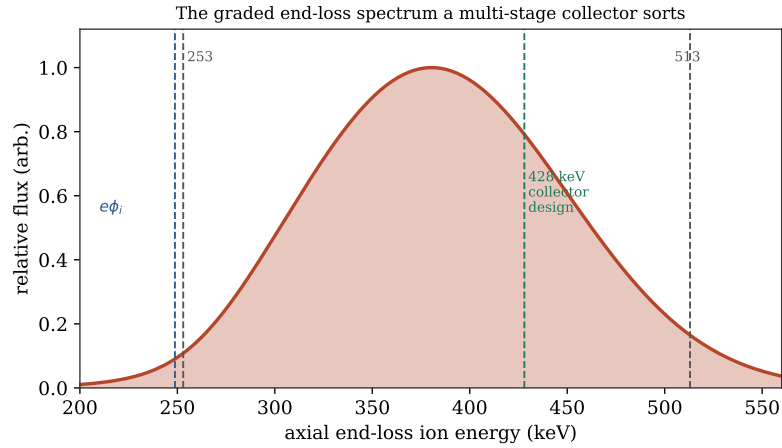


Figure 4. The graded central-cell end-loss ion spectrum (253–513 keV; public-safe window [1]) that a multi-stage collector must sort, referenced to the ambipolar potential $e\phi_i = 248.8$ keV and the 428 keV collector design point (BV-10). (Schematic distribution over register-anchored energy bounds.)

residual $E - eV_k$ as heat. For an exhaust energy distribution $g(E)$ the collector efficiency is

$$\eta_{\text{coll}} = \eta_{\text{grid}} \frac{\sum_{k=1}^N V_k \int_{eV_k}^{eV_{k+1}} g(E) dE}{\int E g(E) dE}, \quad (8)$$

where $\eta_{\text{grid}} = 0.93$ folds in grid transparency, secondary emission and lead losses [2]. Optimizing the boundaries $\{V_k\}$ for the spectrum of figure 4 gives the staircase of figure 5: three stages reach $\eta_{\text{coll}} = 0.70$ and six reach 0.83 (BV-10 [1]), approaching the grid ceiling; a fully graded profile over the real spectrum uses 10–13 stages (MV-CX-27 [1, 35]).

The current the collector can decelerate is bounded by space charge. Across a gap d held at retarding potential V , the maximum decelerated current density is the Child–Langmuir value [18]

$$J_{\text{CL}} = \frac{4\varepsilon_0}{9} \sqrt{\frac{2e}{m}} \frac{V^{3/2}}{d^2}. \quad (9)$$

The raw end-loss current density far exceeds J_{CL} at reactor power, so the exhaust must first be fanned out: the magnetic expander lowers the current density by the flux-expansion ratio R_{exp} before the beam reaches the electrodes, which is precisely what makes electrostatic DEC survivable at power [5, 36]. A Debye-scale cusp at the expander throat additionally sorts the flow by species, sending electrons and ions to different collector banks.

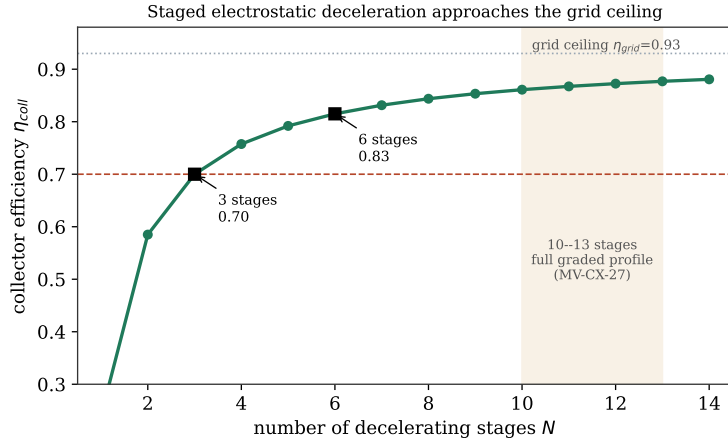


Figure 5. Multi-stage electrostatic collector efficiency, equation (8). Optimized staged deceleration reaches 0.70 at three stages and 0.83 at six (register anchor BV-10), approaching the $\eta_{\text{grid}} = 0.93$ grid ceiling; the fully graded profile uses 10–13 stages.

2.5 The travelling-wave converter and the quasineutral expander

The 14.68 MeV proton channel is too energetic for a compact electrostatic staircase; it is decelerated instead by a *travelling-wave* direct converter (TWDEC), in which a pre-bunched proton beam gives up energy to a travelling electromagnetic wave whose phase velocity is matched to the decelerating beam [19, 20]. The measured tube efficiency of the Kronos travelling-wave stage is the frozen conservative anchor $\eta_{\text{dec}} = 0.49$; against the raw $\approx 34\%$ end-loss energy spread the tube returns ≈ 0.51 , and it only overtakes the electrostatic collector’s 0.70 with an upstream buncher producing a narrow beam (AG-CX-12 [1]).

The directed *electron* channel identified in (7) needs its own converter. We propose the inverse of the usual magnetic-nozzle usage: a quasineutral, two-species expander that recovers the electron channel against the ambipolar potential of the expansion, so that no space-charge wall forms. Along the expanding field the first adiabatic invariant is conserved,

$$\mu = \frac{mv_{\perp}^2}{2B} = \text{const}, \quad \Rightarrow \quad v_{\perp}^2(B) = v_{\perp,0}^2 \frac{B}{B_0}, \quad (10)$$

so as the field drops by the expansion ratio $R_{\text{exp}} = B_{\text{throat}}/B_{\text{end}}$ the perpendicular energy is converted to parallel streaming energy (a magnetic nozzle). In a quasineutral two-species expansion

the electrons set up an ambipolar potential $\Delta\phi \simeq (T_e/e) \ln(R_{\text{exp}})^{1/2}$ that decelerates the electron flow; collecting the electron stream against $\Delta\phi$ rather than letting it charge a wall is the converter. On the adiabatic-invariant backbone of recent expander theory [21], one worked embodiment — $R_{\text{exp}} = 20$, $5 \text{ T} \rightarrow 0.25 \text{ T}$ — recovers a bounded $\approx 28\%$ of the electron channel after modelled demagnetization and detachment losses (figure 6), lifting the axial electric recovery by $\approx 13\%$ ($\approx 97 \text{ MWe}$). This is a bounded upside, not a closure mechanism, and even a bounded-negative experimental result closes a named open problem in advanced-fuel energy conversion.

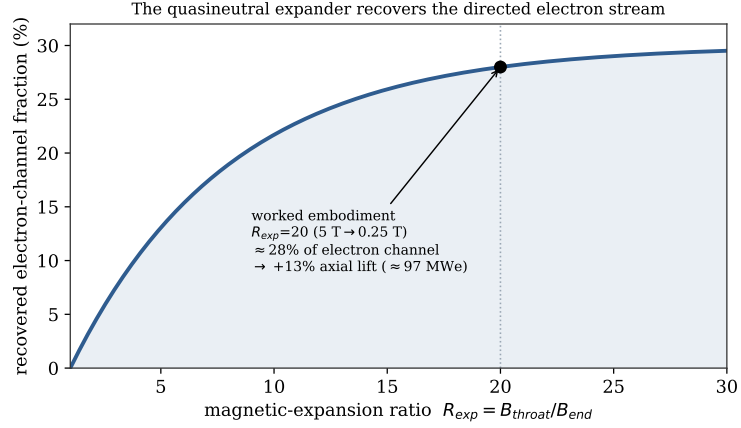


Figure 6. The quasineutral two-species expander, equation (10). Recovered fraction of the directed electron channel versus magnetic-expansion ratio; the worked embodiment ($R_{\text{exp}} = 20$, $5 \text{ T} \rightarrow 0.25 \text{ T}$) returns $\approx 28\%$ after modelled losses, an $\approx 13\%$ / $\approx 97 \text{ MWe}$ axial-recovery lift. Model output, not a measurement.

2.6 Plant power balance and Q_E closure

The converter train (figure 7) delivers a gross electric output

$$P_{\text{out}} = \eta_{\text{dec}} P_{\text{ax}} + \eta_{\text{th}} (P_{\text{exh}} - P_{\text{ax}}), \quad (11)$$

where P_{ax} is the charged axial end-loss power routed to the direct converter, $P_{\text{exh}} - P_{\text{ax}}$ is the radiated and thermalized remainder recovered by a Carnot-bounded bottoming cycle at $\eta_{\text{th}} = 0.45$, and the engineering gain is

$$Q_E = \frac{P_{\text{out}}}{P_{\text{in}}}, \quad P_{\text{in}} = \frac{P_{\text{heat}}}{\eta_{\text{wp}}}, \quad \text{net} = P_{\text{out}} - P_{\text{in}}. \quad (12)$$

At the ratified 440 m design unit, booked at the Moir–Barr design converter efficiency $\eta_{\text{dec}} = 0.70$ (BU-L1-A10 [1]): $P_{\text{ax}} = 2719.7 \text{ MW}$ gives a direct-converted 1903.8 MWe (54.1% of gross), the thermal path adds 1618.2 MWe , $P_{\text{out}} = 3522.0 \text{ MWe}$ against $P_{\text{in}} = 2672.0 \text{ MWe}$, for $Q_E = 1.318$ and net $+850.0 \text{ MWe}$ at recirculating fraction $\varepsilon = P_{\text{in}}/P_{\text{out}} = 0.759$. Two labels must be kept distinct: the charged *fraction of exhaust* $f_{\text{dir}} = P_{\text{ax}}/P_{\text{exh}} = 0.4945$ is not the DEC *share of electricity* 0.541 , and the overall exhaust-to-electric efficiency is $P_{\text{out}}/P_{\text{exh}} = 0.640$ [1]. Section 4 develops the sensitivity of (12) to η_{dec} and shows why the correct reading is that DEC is a bounded lever.

3 Computational methodology (GPU HPC)

Every headline number above is reproduced by a specific code on the Kronos NVIDIA GPU HPC campaign; the fast 0-D and circuit models run on the in-house node so they never contend with the first-principles fleet. No result depends on proprietary data, and no cost or budget quantity enters this paper.

0-D tandem-mirror power balance. The KRONOS toolkit mirror solver (`solve_mirror`, Mode-M) closes equations (3)–(12) with Pastukhov confinement (5), Bremsstrahlung and synchrotron radiation, and the Bosch–Hale reactivity [15]. It reproduces the frozen 440 m anchor byte-for-byte: $Q_E = 1.3181$, $P_{\text{fus}} = 4298.5 \text{ MW}$, $P_n = 233.95 \text{ MW}$, $f_n = 5.44\%$, $e\phi_i = 248.75 \text{ keV}$, self-consistent $T_e = 89.72 \text{ keV}$, $P_{\text{brems}} = 743.10 \text{ MW}$, $P_{\text{syn}} = 1803.04 \text{ MW}$ (BU-L1-A1 [1, 33]).

Kinetic collector and expander physics. The electrostatic-collector deceleration (8), the Child–Langmuir current limit (9), the cusp-expander species sorting and the secondary-emission grid losses are resolved with the particle-in-cell codes WarpX and SMILEI (a 1D3V loss-cone

pipeline established locally, with the full maps as GPU-fleet runs). The travelling-wave stage and the two-species expander recovery (10) are advanced on the same PIC backbone [34, 35].

Slowing-down partition and non-Maxwellian tail. The Stix partition (7) and the DEC-feeding end-loss spectrum are computed with a 1-D Fokker–Planck solver on Rosenbluth potentials [12]; the burner ion distribution is strongly non-Maxwellian, and beam/heating energy is a real knob that enriches the fusion-relevant tail and reshapes the converter-feeding spectrum (AG-CX-04 [1]).

Direct-converter electrode materials. The collector and travelling-wave electrodes are screened for low activation and thermal–mechanical stability with the CHGNet machine-learning interatomic potential and confirmed by Quantum ESPRESSO (PBE PAW) density-functional theory; the down-selected tantalum- and rhenium-free family (e.g. $\text{W}_{48}\text{Ti}_{33}\text{Cr}_{10}\text{Ta}_4\text{V}_4$; the near-neutral $\text{Cr}_2\text{Ti}_5\text{V}_1\text{W}_8$ at formation energy -0.015 eV/atom) is a composition-of-matter result [1, 38]. Parasitic-neutron activation of the electrodes under the $f_n = 5.44\%$ budget is checked with OpenMC. All first-principles work runs on the NVIDIA A100/H100/A10 fleet.

4 Results

4.1 The power-balance chain

Table 1 collects the frozen power-balance chain from the fusion source to the grid, and figure 7 shows the DEC train that realizes it: charged exhaust → cusp expander → graded electrostatic collector → travelling-wave converter → thermal bottoming → HVDC-native output. The direct channel carries 54.1 % of the gross electricity with no thermal cycle; the remainder is recovered thermally.

Table 1. Frozen power-balance chain for the 440 m tandem-mirror burner design unit, booked at the Moir–Barr design converter efficiency $\eta_{\text{dec}} = 0.70$ (BU-L1-A10, BU-L1-A1 [1]). “[T]” denotes a toolkit-computed anchor, “[D]” a derived quantity.

quantity	symbol	value	basis
fusion power	P_{fus}	4298.5 MW	[T]
neutron power / fraction	P_n, f_n	233.95 MW / 5.44 %	[T]
plug potential	$e\phi_i$	248.75 keV	[T]
electron temperature	T_e	89.72 keV	[T]
Bremsstrahlung / synchrotron	$P_{\text{brems}}, P_{\text{syn}}$	743.10 / 1803.04 MW	[T]
charged axial end-loss	P_{ax}	2719.7 MW	[T]
direct-converted electric	$\eta_{\text{dec}} P_{\text{ax}}$	1903.8 MWe (54.1 %)	[T]
thermal-path electric	$\eta_{\text{th}}(P_{\text{exh}} - P_{\text{ax}})$	1618.2 MWe	[D]
gross electric output	P_{out}	3522.0 MWe	[T]
recirculating input	P_{in}	2672.0 MWe	[T]
net electric	net	+850.0 MWe	[T]
recirculating fraction	ε	0.759	[D]
charged fraction of exhaust	f_{dir}	0.4945	[T]
exhaust-to-electric	$P_{\text{out}}/P_{\text{exh}}$	0.640	[D]
engineering gain	Q_E	1.318	[T]

The direct-energy-conversion train (no thermal cycle for the direct channel)

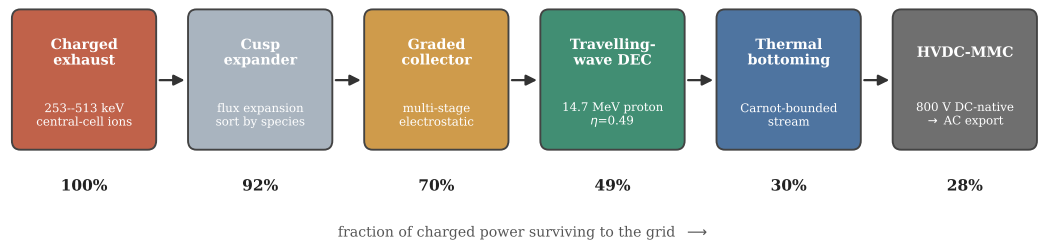


Figure 7. (Schematic.) The direct-energy-conversion train. Central-cell exhaust (253–513 keV) fans out in the cusp expander, is sorted by species, and is decelerated in a graded electrostatic collector; the 14.68 MeV proton channel is handled by the travelling-wave converter at the frozen anchor $\eta_{\text{dec}} = 0.49$; the Carnot-bounded remainder is recovered by a thermal bottoming cycle; the DC output is delivered HVDC-native. Fractions are illustrative of the surviving charged-power cascade, not measured stage yields.

4.2 The direct-converter efficiency ladder

The single most misused number in advanced-fuel DEC is the converter efficiency, because different sources quote different points of the same ladder as if they were one figure. We fix the ladder explicitly (figure 8, table 2): a frozen conservative anchor $\eta_{\text{dec}} = 0.49$ (the measured travelling-wave tube efficiency, with the electron/thermal remainder *not* credited at converter efficiency); a conservative floor 0.56; the Moir–Barr venetian-blind design value 0.70 used to close the power balance; and an ideal proton-only ceiling 0.85 [1, 2]. The frozen anchor sits deliberately *below* the ladder floor, so that the plant case never rests on optimistic conversion (BU-SX-01, ADJ3 [1]).

Table 2. The DEC efficiency ladder (BU-SX-01/03, ADJ3, BV-10, AG-CX-12 [1]). The frozen canonical anchor is the conservative measured travelling-wave tube efficiency; the power balance of table 1 is closed at the Moir–Barr design value.

rung	η_{dec}	basis
frozen conservative anchor	0.49	measured travelling-wave tube efficiency
travelling-wave, raw spread	0.51	TWDEC over $\approx 34\%$ end-loss spread (AG-CX-12)
conservative floor	0.56	pessimistic collector bound
Moir–Barr design	0.70	venetian-blind / graded electrostatic (power-balance basis)
ideal proton-only ceiling	0.85	line-matched proton-only limit

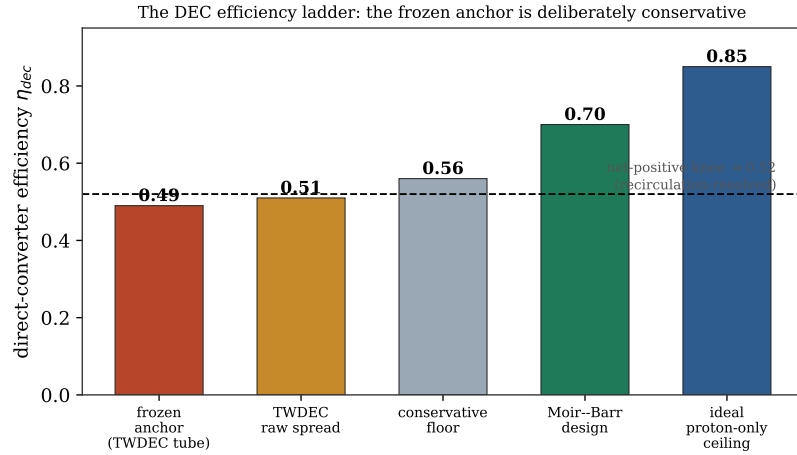


Figure 8. The DEC efficiency ladder. The frozen canonical anchor (0.49) is deliberately conservative and sits below both the ladder floor (0.56) and the recirculation-resolved net-positive knee (≈ 0.52); the power balance of table 1 is closed at the Moir–Barr design value (0.70), well below the ideal ceiling (0.85).

4.3 Conversion is a bounded lever, not the closure mechanism

Substituting (11) into (12) makes Q_E and net electric explicit functions of η_{dec} (figure 9a). In the gross balance, net electric passes through +850 MWe at the design value 0.70; resolving the recirculating plug loop (companion [33]) places the operational net-positive knee at $\eta_{\text{dec}} \approx 0.52$ — just above the frozen conservative anchor 0.49. The reading is unambiguous: because the net knee is so close to the design point, *converter efficiency is a bounded lever around closure, not the mechanism that produces it*. What actually buys net margin is central-cell length: the length ladder gives net +104.2, +850.0 and +2831.9 MWe at 55, 440 and 1400 m (figure 9b), because lengthening the cell raises fusion power against a fixed plug and radiation overhead. The honest gate on the generator is the recirculating plug power, and DEC improves the generator without being what makes it close — which is exactly why a bounded-negative experimental result on the converter would not threaten the machine.

4.4 Product fit: Aegis and MetroVolt

Because the converter output is native direct current, the two burner products exploit it differently (table 3, figure 10). *MetroVolt*, the behind-the-meter data-center burner, delivers HVDC-native power to the racks: a two-stage DEC-HVDC $\rightarrow$ MVDC (± 10 kV) $\rightarrow$ 800 V path at $\approx 98.2\%$ conversion (MV-CX-01), giving 88.1% source-to-chip against 86.2% for an OCP-DC chain and 75.9% for a conventional AC chain — a +12.2-point gain that deletes the double-conversion UPS

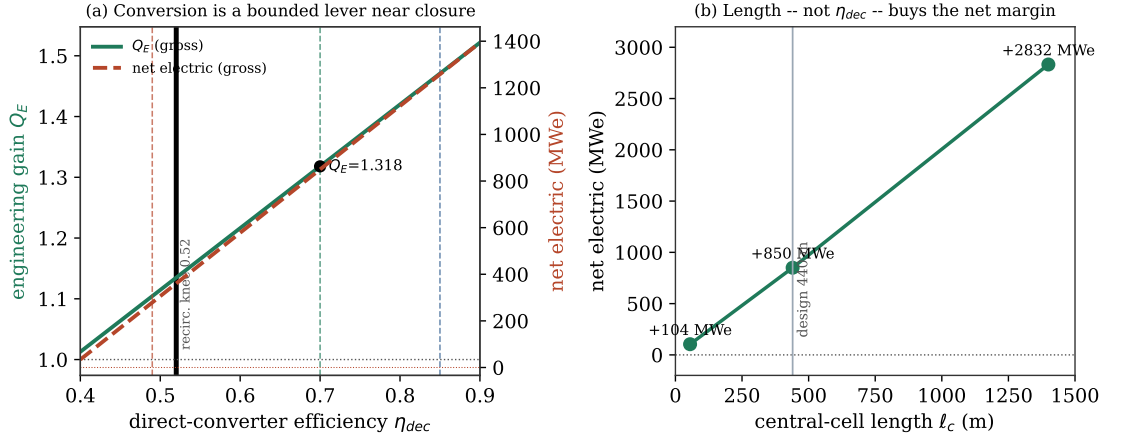


Figure 9. DEC is a bounded lever. (a) Q_E (left axis) and net electric (right axis) versus converter efficiency, equations (11)–(12): the design point closes at $Q_E = 1.318$ / +850 MWe ($\eta_{dec} = 0.70$); the recirculation-resolved net knee (≈ 0.52) sits just above the frozen anchor (0.49). (b) The length ladder: net electric versus central-cell length — length, not η_{dec} , buys the margin. All points are register anchors [1].

and the plant generator/step-up chain (MV-CX-02 [1, 37]). The direct-DC bus is inherently smooth (1 % ripple needs only $\approx 234 \mu\text{F}$; the ITIC hold-up requirement dominates) and current-limited at source ($\approx 3 \text{ kA}$ fault against $\approx 50 \text{ kA}$ from a grid tie), a safety advantage a SiC solid-state breaker exploits (MV-CX-03/05). *Aegis*, the forward-base burner, exploits the same directness for grid independence: on a $\approx 3 \text{ MWh}$ stored-energy store the DEC output crosses the start-up draw at $\approx 42 \text{ s}$ from cold (HX-07), and millisecond DEC trimming with an $\approx 8 \text{ s}$ plasma lag is bridged by a small $\approx 246 \text{ MWe}$ / 1.1 MWh buffer for islanded load-following (AG-CX-20) [1, 30].

Table 3. Direct-conversion product fit. Source-to-chip and black-start figures are register anchors (MV-CX-01/02/03/05, HX-07, AG-CX-20 [1]).

attribute	MetroVolt (data-center)	Aegis (forward-base)
delivery	HVDC-native to racks	islanded microgrid
source-to-chip / chain	88.1 % vs 75.9 % AC	grid-independent
conversion stages	2–3 (DEC direct)	2–3 (DEC direct)
black-start	—	cross-over $\approx 42 \text{ s}$ / $\approx 3 \text{ MWh}$
load-following	ms DEC trim	ms DEC trim, 246 MWe/1.1 MWh buffer
fault current at source	$\approx 3 \text{ kA}$ (50 kA grid)	current-limited

4.5 Verification status

Table 4 states each DEC claim as a target or a reproduced register result with its serial. The power-balance decomposition and the efficiency ladder are reproduced anchors; the converter itself is not built, and the recovered-fraction figures are worked-embodiment model outputs, marked as such.

Table 4. DEC verification status. “reproduced” denotes an independently re-run register anchor; “model” denotes a worked-embodiment output; “target” a pre-registered bar. Serials refer to the register [1].

claim	value	serial	status
power-balance closure	$Q_E = 1.318$, net +850 MWe	BU-L1-A10	reproduced
charged axial end-loss	$P_{ax} = 2719.7 \text{ MW}$	BU-L1-A10	reproduced
collector staging	0.70 (3 st.) / 0.83 (6 st.)	BV-10	reproduced
efficiency ladder	0.49/0.56/0.70/0.85	BU-SX-01, ADJ3	reproduced
net-positive knee	$\eta_{dec} \approx 0.52$	BU-L1-A10	reproduced
electron-dominant axial split	55.5/44.5 %	register anchor	model
expander electron recovery	$\approx 28 \%$ (+97 MWe)	source embodiment	model
TWDEC over raw spread	≈ 0.51	AG-CX-12	reproduced
MetroVolt source-to-chip	88.1 % vs 75.9 % AC	MV-CX-02	reproduced
Aegis black-start	cross-over $\approx 42 \text{ s}$	HX-07	reproduced
low-activation electrodes	Ta/Re-free, -0.015 eV/at	BR-L2-A37c	reproduced

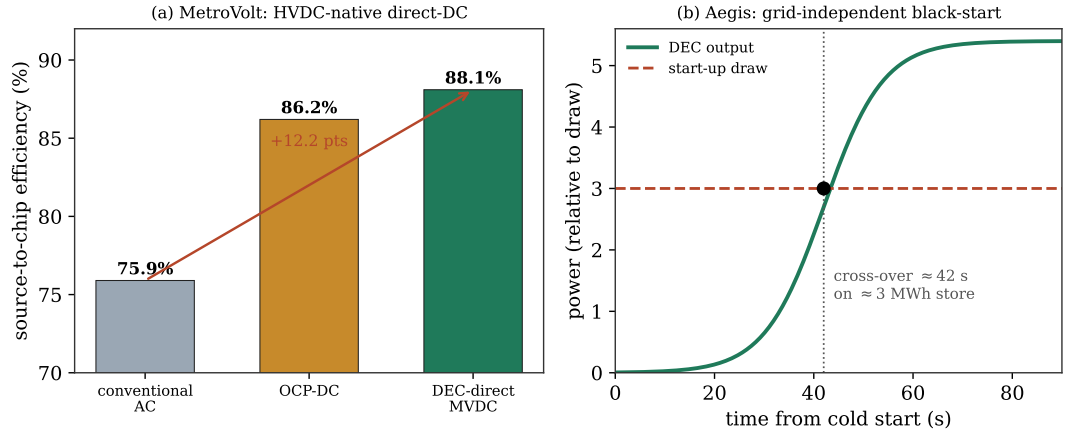


Figure 10. Product fit of native-DC direct conversion. **(a)** MetroVolt source-to-chip efficiency: DEC-direct MVDC (88.1%) against OCP-DC (86.2%) and a conventional AC chain (75.9%), a +12.2-point gain (MV-CX-02). **(b)** Aegis grid-independent black-start: DEC output crosses the start-up draw at ≈ 42 s on a ≈ 3 MWh store (HX-07). Register anchors [1].

5 Discussion

5.1 Against the venetian-blind and travelling-wave literature

The Kronos design converter efficiency (0.70) is exactly the value Moir and Barr established for the venetian-blind converter in the 1970s and confirmed in beam-direct-converter experiments [2, 3]; our contribution is not a higher number but a more honest one — freezing the plant-level anchor at 0.49, below the ladder floor, and showing the closure does not depend on the difference. The travelling-wave stage follows the GAMMA-10 line of TWDEC experiments [20, 19], which demonstrated deceleration of real mirror end-loss ions; our 0.49 tube anchor and ≈ 0.51 raw-spread figure are conservative against those results and rise only with an upstream buncher. Placing DEC in a recirculating balance also situates it against Rider’s fundamental limits on advanced-fuel systems near thermal equilibrium [16, 17]: because the burner recirculates a large fraction ($\varepsilon = 0.759$) to sustain the plug, the plant is honestly gated by that recirculation, and a converter improvement is a bounded lever rather than an escape from Rider’s accounting.

5.2 The directed-electron reframing

The scientific novelty is the relocation of the DEC target. The standard framing follows the 14.68 MeV proton because it carries the birth energy; the slowing-down partition (7) shows that at reactor temperature that proton has already handed most of its energy to the electrons before it can be collected, so the dominant axial channel is a directed electron stream that no published converter recovers [5, 2]. The quasineutral two-species expander of §2.5 is the inverse of the magnetic nozzle that mirror end-cells already use as a thruster [21, 36]; running it backward, as a collector against the ambipolar potential, is what recovers the electron channel. This is a named, testable open problem, and the modest $\approx 13\%$ recovery it promises is scoped honestly as upside.

5.3 In the context of the modern mirror revival

The burner rides a genuine revival of the axisymmetric mirror. High-field HTS mirrors (WHAM) now have a published physics basis [22], the gas-dynamic trap has demonstrated the electron-temperature and ambipolar-potential physics that the plug relies on [23, 24], and simpler tandem configurations have been re-examined [9, 8]. Direct conversion is the piece of the advanced-fuel mirror that has *not* been re-derisked at reactor scale in that revival, and specifying it against a real burner power balance — rather than a fuel-cycle sketch [19, 25, 26] — is where this paper sits. The two-magnet discipline (the 26.49 T plug distinct from the 16.84 T breeder centrepost [32, 1]) and the microstability constraints on the plug [10] are carried by companion papers; here they enter only as the boundary condition that sets $e\phi_i$ and the end-loss spectrum.

6 Limitations

Kronos has built no converter; every number here is a design target, a reproduced register anchor, or an open-risk model inference. The 0.49 anchor is a conservative measured travelling-wave tube efficiency, not a full-system plant measurement; the $\approx 28\%$ expander electron recovery and the

55.5/44.5 axial split are worked-embodiment and model outputs, not measurements, and the split rests on booking the collisional partition against a finite confinement time (5). The space-charge and secondary-emission physics of a reactor-scale collector, and the neutron and thermal fatigue of the electrodes, are pre-registered follow-on items (AG-CX-09/10 [1]). None of these bear on the generator's closure, which the plug gate — not the converter — governs; that separation is the paper's load-bearing honest statement.

7 Conclusion

For a D-³He tandem-mirror burner the exhaust is overwhelmingly charged ($f_n = 5.44\%$), so its power can be recovered directly, without a thermal cycle on the direct channel. Writing the physics out, the axial loss at reactor temperature is dominated by a directed *electron* stream ($\approx 55.5\%$ against $\approx 44.5\%$ ions), inverting the birth-spectrum picture across 20–200 keV, and no published converter recovers it; a quasineutral two-species expander is the inverse of the standard nozzle and returns a bounded $\approx 28\%$ of that channel. The converter efficiency is a ladder — frozen conservative anchor 0.49, design 0.70, ceiling 0.85 — and, placed in the recirculating balance, DEC is a bounded upside lever: the burner closes at $Q_E = 1.318$ and net +850.0 MWe, the net knee sits at $\eta_{\text{dec}} \approx 0.52$, and closure is gated by the recirculating plug power, not by conversion. The direct, native-DC output is the product edge that gives MetroVolt an 88.1% HVDC-native source-to-chip path and Aegis a ≈ 42 s grid-independent black-start. The reframing names and scopes an open problem in advanced-fuel energy conversion, and does so without letting the plant lean on it.

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Reproducibility. Every headline quantity in this paper is a frozen anchor in the KRONOS Physics De-Risking Register^[R0] (doi:10.5281/zenodo.22133057) and is regenerable from the archived toolkit and the uniquely named figure-generation script (`make_dec_figs_kfe14.py`) deposited with this paper; this paper is archived at doi:10.5281/zenodo.22645700; official version: <https://www.kronosfusionenergy.com/publications/direct-energy-conversion.html>. The power-balance chain (BU-L1-A10, BU-L1-A1), the efficiency ladder (BU-SX-01/03, ADJ3, BV-10), the collector staging (BV-10), and the product-fit figures (MV-CX-02, HX-07) are the load-bearing anchors. No result depends on unpublished or proprietary data, and no economic quantity appears.

Scope — what this paper does not claim. Kronos has built no converter; the recovered-fraction figures are worked-embodiment model outputs after modelled losses, not measurements, and direct energy conversion is a bounded upside lever, not the machine's closure mechanism.

Data availability

This paper is archived at doi:10.5281/zenodo.22645700. Headline quantities are frozen anchors in the register [1] and cross the series: the Aegis burner [30], the MetroVolt burner [31], the REBCO plug and centrepost magnet basis [32], the recirculation-resolved Q_E closure [33], the collector-physics and converter-architecture studies [34, 35], the end-cell/expander loss budget [36], the HVDC-native load interface [37], and the low-activation electrode family [38]. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

References

- [1] P I Ford, *KRONOS Physics De-Risking Register*, v8, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22133057
- [2] R W Moir and W L Barr, Venetian-blind direct energy converter for fusion reactors, *Nucl. Fusion* **13** 35 (1973). doi:10.1088/0029-5515/13/1/005
- [3] W L Barr and R W Moir, Test results on plasma direct converters, *Nucl. Technol./Fusion* **3** 98 (1983).
- [4] R F Post, Mirror systems: fuel cycles, loss reduction and energy recovery, in *Nuclear Fusion Reactors* (British Nuclear Energy Society, London, 1970) p. 88.
- [5] R F Post, The magnetic mirror approach to fusion, *Nucl. Fusion* **27** 1579 (1987). doi:10.1088/0029-5515/27/10/001
- [6] V P Pastukhov, Collisional losses of electrons from an adiabatic trap in a plasma with a positive potential, *Nucl. Fusion* **14** 3 (1974). doi:10.1088/0029-5515/14/1/001
- [7] R H Cohen, M E Rensink, T A Cutler and A A Mirin, Collisional loss of electrostatically confined species in a magnetic mirror, *Nucl. Fusion* **18** 1229 (1978). doi:10.1088/0029-5515/18/9/005
- [8] D E Baldwin and B G Logan, Improved tandem mirror fusion reactor, *Phys. Rev. Lett.* **43** 1318 (1979). doi:10.1103/PhysRevLett.43.1318
- [9] T K Fowler, R W Moir and T C Simonen, A new simpler way to obtain high fusion power gain in tandem mirrors, *Nucl. Fusion* **57** 056014 (2017). doi:10.1088/1741-4326/aa5e54
- [10] R F Post and M N Rosenbluth, Electrostatic instabilities in finite mirror-confined plasmas, *Phys. Fluids* **9** 730 (1966). doi:10.1063/1.1761740
- [11] T H Stix, Heating of toroidal plasmas by neutral injection, *Plasma Phys.* **14** 367 (1972). doi:10.1088/0032-1028/14/4/002
- [12] M N Rosenbluth, W M MacDonald and D L Judd, Fokker–Planck equation for an inverse-square force, *Phys. Rev.* **107** 1 (1957). doi:10.1103/PhysRev.107.1
- [13] B A Trubnikov, Particle interactions in a fully ionized plasma, *Rev. Plasma Phys.* **1** 105 (1965).
- [14] L Spitzer, *Physics of Fully Ionized Gases*, 2nd ed. (Interscience, New York, 1962).
- [15] H-S Bosch and G M Hale, Improved formulas for fusion cross-sections and thermal reactivities, *Nucl. Fusion* **32** 611 (1992). doi:10.1088/0029-5515/32/4/I07
- [16] T H Rider, Fundamental limitations on plasma fusion systems not in thermodynamic equilibrium, *Phys. Plasmas* **4** 1039 (1997). doi:10.1063/1.872556
- [17] W M Nevins, A review of confinement requirements for advanced fuels, *J. Fusion Energy* **17** 25 (1998). doi:10.1023/A:1022513215080
- [18] I Langmuir, The effect of space charge and residual gases on thermionic currents in high vacuum, *Phys. Rev.* **2** 450 (1913). doi:10.1103/PhysRev.2.450
- [19] H Momota *et al*, Conceptual design of the D–³He reactor ARTEMIS, *Fusion Technol.* **21** 2307 (1992).
- [20] Y Yasaka, H Takeno *et al*, Study of direct energy conversion of mirror end-loss plasma for advanced-fuel fusion (GAMMA-10 travelling-wave direct energy converter), *Fusion Sci. Technol.* **51** 143 (2007).
- [21] J-M Rax, E J Kolmes and N J Fisch, Efficiency of direct energy conversion in expanding plasmas (2025).
- [22] D Endrizzi *et al*, Physics basis for the Wisconsin HTS Axisymmetric Mirror (WHAM), *J. Plasma Phys.* **89** 975890501 (2023). doi:10.1017/S0022377823000806
- [23] P A Bagryansky *et al*, Threefold increase of the bulk electron temperature of plasma discharges in a magnetic mirror device, *Phys. Rev. Lett.* **114** 205001 (2015). doi:10.1103/PhysRevLett.114.205001
- [24] A A Ivanov and V V Prikhodko, Gas-dynamic trap: an overview of the concept and experimental results, *Plasma Phys. Control. Fusion* **55** 063001 (2013). doi:10.1088/0741-3335/55/6/063001
- [25] L J Wittenberg, J F Santarius and G L Kulcinski, Lunar source of ³He for commercial fusion power, *Fusion Technol.* **10** 167 (1986).
- [26] G A Wurden *et al*, A new vision for fusion energy research: fusion rocket engines for planetary defense, *J. Fusion Energy* **35** 123 (2016). doi:10.1007/s10894-015-0034-1
- [27] T C Simonen, Experimental progress in magnetic-mirror fusion research, *Proc. IEEE* **69** 935 (1981).
- [28] J Wesson, *Tokamaks*, 4th ed. (Oxford University Press, Oxford, 2011).
- [29] J D Huba, *NRL Plasma Formulary* (Naval Research Laboratory, Washington, DC, 2018).

- [30] P I Ford and G L Kulcinski, The Aegis forward-base burner: Q_E closure, plug requirement, and mirror confinement, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645693
- [31] P I Ford and G L Kulcinski, The MetroVolt data-center burner: direct-DC campus power and the plug requirement, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645695
- [32] P I Ford, R J Weggel and C Weggel, The REBCO high-field superconductor basis for the breeder centrepost and burner plug, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645697
- [33] P I Ford, Recirculation-resolved Q_E closure in a D-³He tandem-mirror burner: the synchrotron-limited operating point and the carried-open plug gate, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645860
- [34] P I Ford, Direct-energy-converter collector physics in a D-³He tandem-mirror burner: multi-stage deceleration, secondary emission, and the space-charge-limited grid optimum, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645858
- [35] P I Ford, Direct-energy-converter architectures for a D-³He tandem-mirror burner: venetian-blind, traveling-wave, and periodic-electrostatic-plug collectors compared, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645935
- [36] P I Ford, End-cell, expander, and halo physics in a D-³He tandem-mirror burner: the mirror ends that set the loss budget, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645852
- [37] P I Ford, The direct-energy-converter-to-load interface in a D-³He tandem-mirror data-center burner: an HVDC-native power-electronics path with millisecond load-following, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645783
- [38] P I Ford, A tantalum- and rhenium-free low-activation alloy family for first wall, divertor and direct-energy-converter electrodes, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645891
- [39] P I Ford, Tandem-mirror design-space mapping for a D-³He burner: 0-D power balance benchmarked against WHAM and the Pastukhov-plug operating point, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645911